SOUNABHA MAJUMDAR

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Codeforces | CodeChef | LeetCode

Education

K. J. Somaiya School of Engineering, Mumbai, B.Tech in Computer Engineering, CGPA: 9.53 — 2024 - 2028
Pace Junior Science College, Chembur, HSC (Maharashtra State Board), 81.5% — 2024
Atomic Energy Central School No. 2, SSC (CBSE), 92.4% — 2022

Experience

LLMOps Intern, Euclid, K. J. Somaiya School of Engineering — June 2026 - July 2026

- Authored the team's **LLMOps lifecycle framework** covering prompt management, RAG data preparation, PEFT (LoRA, adapters, prefix tuning), evaluation pipelines, monitoring and serving with **vLLM**, used as the reference for its SLM work
- Analyzed operational challenges in **LLM evaluation and cost**, from hallucination detection and the limits of n-gram metrics to LLM-as-a-judge, DeepEval and cost versus latency, ranking them by impact and maturity to prioritize SLM development
- Wrote a production **LLM security guide** on threat modeling, trust boundaries and direct vs indirect **prompt injection**, comparing guardrails (Llama Guard, Prompt Guard, ShieldGemma, NeMo Guardrails, Granite Guardian) by pipeline position

Projects

Distributed S3-Compatible Object Store — GitHub

- Built a **distributed object store in Go** (stateless gateway, **3-node Raft** metadata cluster on BoltDB, 6 storage nodes) that speaks the **S3 API**, including multipart uploads, range reads and AWS SigV4 auth, and works **unmodified with the aws CLI and SDKs**
- Split objects into 4 MiB content-addressed chunks, erasure coded with **Reed-Solomon (4+2)** and placed by a **consistent-hash ring** with 128 virtual nodes, surviving **any 2 node failures** at **1.5x storage overhead** versus 3.0x for replication
- Made writes **crash-safe** with fsync-before-commit ordering and **linearizable** Raft metadata with idempotent retries, plus SHA-256 verified reads, background **repair** of missing or corrupt shards, and mark-and-sweep **garbage collection**
- Chaos-tested with node kills, network partitions, leader loss and 40 corrupted shards: **5,980 acknowledged writes, 0 lost** or corrupted; measured **408 MB/s GET** throughput and a **16 s rebuild** of a wiped node, and fixed a degraded-read bug the benchmarks exposed (**2.9 to 227.5 MB/s**)

Tech: Go, Raft, BoltDB, Reed-Solomon Erasure Coding, Consistent Hashing, Docker, Prometheus, AWS SigV4

AI Workflow Automation Tool (n8n-style) — GitHub

- Built a **visual builder for AI workflows**: users drag HTTP, Gemini LLM, Switch and wait-for-input blocks onto a React Flow canvas and run them manually, on cron, or via signed webhooks, with live status over **WebSockets**
- Ran each block as its own **Celery task on Redis**, with PostgreSQL dependency counters (**Kahn's algorithm**) dispatching a block the moment its inputs are ready and **row-level locks** (SELECT FOR UPDATE) ensuring **no block runs twice**
- Made runs **survive worker crashes** with retryable vs fatal error classification, **exponential backoff** with full jitter, and sweepers that recover stuck or timed-out blocks; Fernet-encrypted credentials, **about 120 pytest tests** and **CI** with an integration smoke test

Tech: Python, FastAPI, Celery, Redis, PostgreSQL, React, React Flow, WebSockets, Docker

Codebase Intelligence Platform (Graph RAG over Code) — In Progress

- Built a platform that indexes repositories into a **Neo4j code graph** using a sandboxed Python indexer, with **SCIP** for precise call resolution (TypeScript, Python, Go) and tree-sitter heuristics for other languages, tested on Hono, Flask, Gin and PocketBase
- Built **cited code Q&A** as a rewrite, retrieve, write and verify pipeline orchestrated in **LangGraph** on Gemini, combining **keyword and pgvector search** with RRF fusion, **cross-encoder reranking** and call-graph expansion, and a **verifier that drops unsupported sentences**
- Added **call-flow tracing** from entry points and **diff impact analysis** via reverse reachability over the graph, served by TypeScript services (Fastify, BullMQ, Next.js) with CI/CD on GitHub Actions

Tech: TypeScript, Python, LangChain, LangGraph, Neo4j, PostgreSQL (pgvector), Redis, tree-sitter, SCIP, Gemini API, RAG

Skills

- **Languages:** C++, Java, Python, Go, JavaScript, TypeScript, SQL, C
- **Backend and Distributed Systems:** FastAPI, Node.js, Express.js, Celery, REST APIs, WebSockets, JWT Auth, Raft, Erasure Coding, Consistent Hashing
- **Databases:** PostgreSQL, MySQL, MongoDB, Redis, Neo4j, BoltDB
- **AI and ML:** PyTorch, Hugging Face, LLM APIs, LangChain, LangGraph, AI Agents, RAG, Vector Databases, LoRA/PEFT
- **Tools and Fundamentals:** Git, Docker, Docker Compose, GitHub Actions CI, pytest, Prometheus, Linux, DSA, OS, DBMS

Achievements

- **ICPC Asia Regionalist 2026** with team O(1)hope, the top-ranked team from my college in the India preliminary round
- **Top 10 finalist**, **Paytm Build for India AI Hackathon** (Mumbai, 2026), building Dukaan Saathi, an AI assistant for shopkeepers
- **3-Star on CodeChef** and **Specialist on Codeforces**, with **1000+** DSA problems solved across platforms
- Ranked **Top 500 among 100,000+** participants nationwide in AlgoUniversity Tech Fellowship selections